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Introduction

This comprehensive report delves deeper into the pressing environmental issues of air quality and CO₂ emissions within the United States, scrutinizing their significant impacts on public health and the broader ecosystem. It methodically identifies and analyzes major sources of pollution and greenhouse gas emissions, evaluating their distribution across the nation to provide tangible insights for policymakers, environmental agencies, and the general public. The study's primary aim is to spotlight severely polluted regions, gauge the efficacy of existing environmental policies, and suggest robust, strategic interventions to mitigate these critical challenges. Furthermore, this analysis addresses the escalating problem of deteriorating air quality and increasing CO₂ emissions, which are crucial for safeguarding public health, ensuring environmental sustainability, and combating climate change. By offering a clearer understanding of the prevailing issues and their ramifications, the report empowers stakeholders to make informed decisions, ultimately steering the nation toward improved environmental health and sustainable living outcomes.

Hypothesis

Areas with higher population densities in the USA exhibit significantly higher greenhouse gas emissions as compared to areas with lower population densities. In those areas, energy use and production are the main sources of greenhouse gas emissions, indicating a direct correlation between energy activities and emission levels.

Data Acquisition and Processing

The first of the data that I acquired was from Worldwide Air Quality's API ([API documentation](#)). Using the latitude and longitude range of the entire United States, I retrieved the station's air quality index from everywhere within the range of the coordinations given, in an attempt to retrieve the AQI (air quality index) of everywhere within the country. While it didn't get as much as I wanted, it still had enough data for me to sufficiently gauge how it changed within the country. After acquiring the data, I found that the station names weren't put in a usable city and state format. To solve this, I used a reverse geocoding API to take the latitude and longitude data and convert it to their respective state and city which I could use for scraping other websites. ([Reverse Geocode API](#))

	Station	Latitude	Longitude	AQI		City	State
0	Sorel-Tracy - secteur de Saint-Joseph-de-Sorel...	46.04	-73.11	20		Sorel-Tracy	Quebec
1	Las Cruces ED, New Mexico, USA	32.31	-106.75	136		Las Cruces	New Mexico
2	Santa Clarita-Placerita, Los Angeles	34.38	-118.53	34		Santa Clarita	California
3	St. Michael, Minnesota, USA	45.21	-93.67	30		Saint Michael	Minnesota
4	Millington, Maryland, USA	39.31	-75.80	30	Millington Wildlife Management Area		Maryland
...	...	...	...	...		...	...
789	Bishop-White Mountain Research Center, Inyo, C...	37.36	-118.33	29		Bishop	California
790	E 1st St, Colville, Washington, USA	48.54	-117.90	55		Colville	Washington
791	Sandpoint, Idaho, USA	48.29	-116.56	28		Ponderay	Idaho
792	Lima, Ohio, USA	40.77	-84.05	24		Township of Bath	Ohio
793	Kenner site, Louisiana, USA	30.04	-90.27	28		District 5	Louisiana

Next, I scraped data from OECD Data Explorer ([Website Data Link](#)). I used Selenium to scrape this site because I found that the data was returned in Javascript and not in HTML. However, with this website specifically, it loaded slowly so I had to allocate more time for it to load and parse the information on the site. Since the data was already in a table format, I just needed to scrape the data and clean it so that I could make visualizations with it. The main issue with most of the data I scraped was that it was being returned as an object and not a float, so it was just mostly changing it back. Then,

I separated the data frame into 4 with each one representing different factors that contribute to the gas emissions.

	Time period	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Average
0	Tonnes of CO2-equivalent, Thousands	352006.12	366975.72	357244.47	357212.67	368665.97	363910.56	357575.72	359082.41	362175.24	366822.71	363206.02	376382.97	362605.05
1	Percentage of greenhouse gas emissions	4.99	5.31	5.36	5.22	5.34	5.40	5.44	5.47	5.36	5.54	6.03	5.94	5.45

	Time period	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Average
0	Tonnes of CO2-equivalent, Thousands	590985.63	582369.36	572600.86	597546.16	603008.19	604944.41	601754.03	613096.61	629494.02	614462.11	597264.51	598099.06	600468.75
1	Percentage of greenhouse gas emissions	8.37	8.43	8.58	8.73	8.74	8.98	9.15	9.34	9.32	9.28	9.91	9.43	9.02

	Time period	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Average
0	Tonnes of CO2-equivalent, Thousands	183135.34	174127.25	176564.79	172827.92	172734.78	171696.30	168325.54	170905.25	173705.11	175993.61	171523.12	169162.94	173391.83
1	Percentage of greenhouse gas emissions	2.59	2.52	2.65	2.53	2.50	2.55	2.56	2.60	2.57	2.66	2.85	2.67	2.60

	Time period	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Average
0	Tonnes of CO2-equivalent, Thousands	5932070.82	5783730.90	5564123.41	5714074.35	5754116.66	5596807.36	5450777.08	5418740.17	5589457.28	5460638.45	4893979.96	5196583.32	5529591.65
1	Percentage of greenhouse gas emissions	84.05	83.73	83.41	83.52	83.41	83.07	82.86	82.58	82.75	82.51	81.21	81.96	82.92

Lastly, using the state and city obtained from the first data frame, I combined it with this website: (<https://dataportalforcities.org/north-america/united-states>). This website has the GHG emissions and energy consumption of each city in the US, and by finding out the URL format (united-states/state/city), I could use the cities and states I obtained from the previous data in collaboration with this URL. I did it this way because the site has 8,000 cities and I didn't want to overload the server with too many requests, so I limited it to the cities I had data on already and focused on gathering data for those. Also, the challenge and reason I opted to do the method above was that all the links were within one page, but needed a button to load 10 more, which I concluded would be too complicated and time-consuming to process. However, another challenge came up when dealing with this data as much of the information was retrieved back in 2015, meaning collaborating this data with more recent data from other websites might lead to inaccuracy.

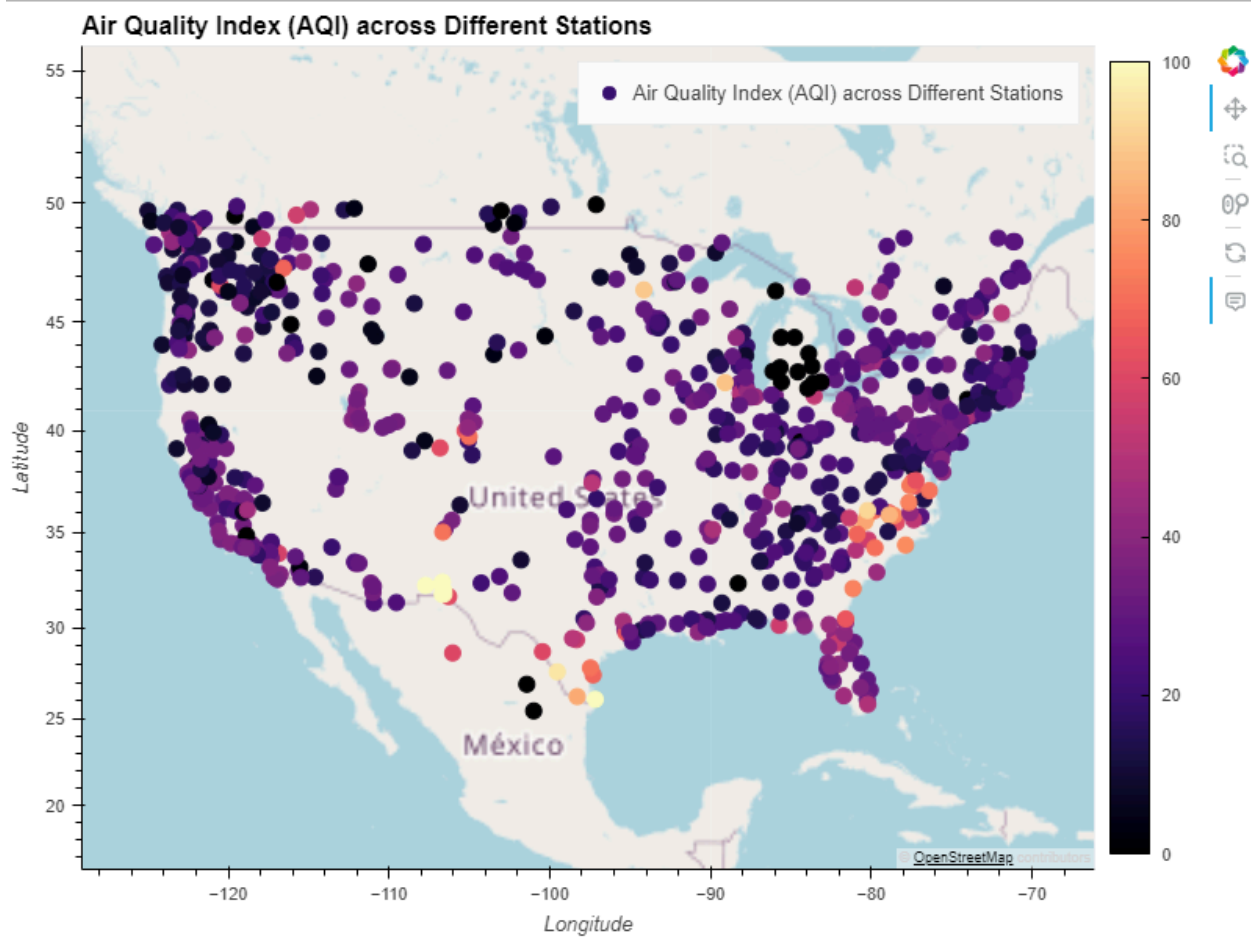
	City	State	Population	Energy Consumption	GHG Emissions
0	Saint Michael	Minnesota	17295	2960.15	214096.78
1	Pocomoke City	Maryland	4101	684.15	51801.40
2	Santa Clarita	California	181972	23580.75	1670902.12
3	Las Cruces	New Mexico	101759	11776.79	1067589.81
4	Suffolk	Virginia	89273	16730.13	1110749.33
...	...	...	...	...	...
437	Roanoke Rapids	North Carolina	15171	1370.46	142457.81
438	Bishop	California	3782	379.51	32405.64
439	Thousand Oaks	California	128888	26134.35	1507604.68
440	Ponderay	Idaho	1155	704.28	52999.49
441	Colville	Washington	4788	371.92	44102.93

Using the data frame above, I also grouped it by its state to create visualizations based on the overall data of the states. The data included 46 states, with some being removed due to cleaning and it being missing from the latitude and longitude coordinates.

	State	Population	Energy Consumption	GHG Emissions
0	Alabama	926489	225200.51	21181458.02
1	Arizona	3895258	552758.09	48807451.73
2	Arkansas	277098	67401.24	5283618.67
3	California	8870813	896774.36	57834307.08
4	Colorado	1597633	274950.73	25369080.44
5	Connecticut	529885	71120.63	4679381.27
6	Delaware	7736	1169.30	87052.97
7	Florida	1558334	314917.00	25206340.22
8	Georgia	499171	56604.68	4458956.53
9	Idaho	241515	24831.91	2205337.79
10	Illinois	773480	143299.44	12906926.26
11	Indiana	125364	27159.65	2351212.30
12	Iowa	671947	122738.33	11513899.07
13	Kansas	525849	103724.79	10213870.31
14	Kentucky	99459	22873.65	2269681.40
15	Louisiana	69677	16766.80	1479938.61
16	Maine	105674	23684.93	1510919.56
17	Maryland	20162	2340.34	206427.24
18	Massachusetts	565053	72362.93	4885752.48
19	Michigan	561251	88122.18	8308080.11
20	Minnesota	984855	174768.37	15699720.67

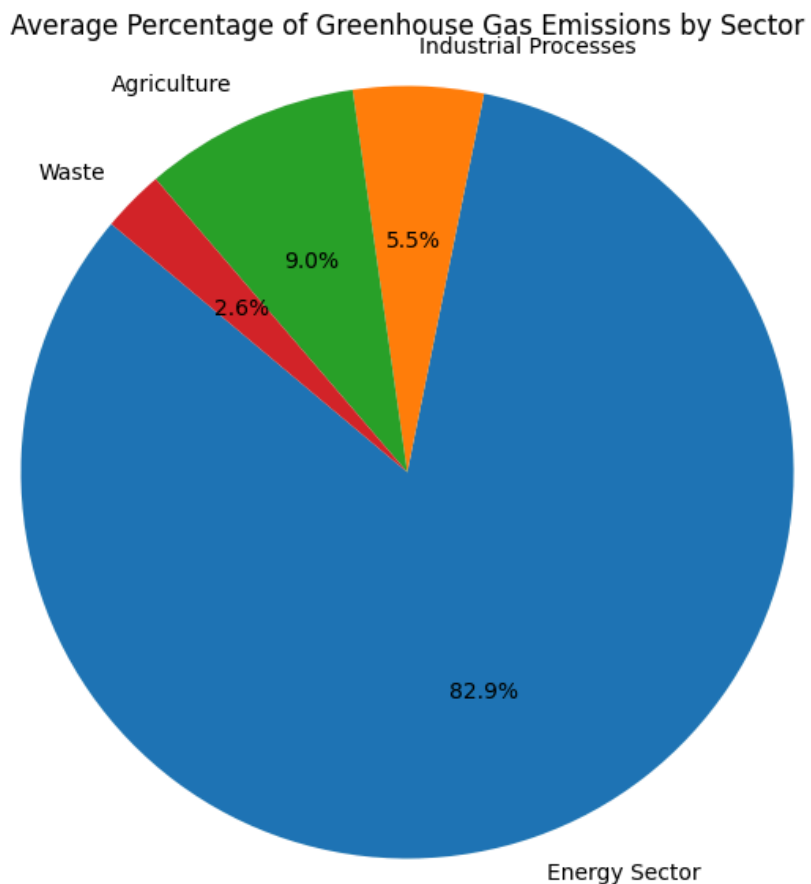
Visualization and Analysis

The map visualizes the Air Quality Index (AQI) across various monitoring stations in the United States. The color gradient from purple to orange indicates AQI levels from low (healthier) to high (unhealthier). Darker purple spots, which are predominant, suggest areas with better air quality. In contrast, areas represented by orange or red dots, though fewer, indicate higher pollution levels. The distribution shows a wide variation in air quality, with certain regions, especially those densely populated or industrial, possibly experiencing worse air quality.



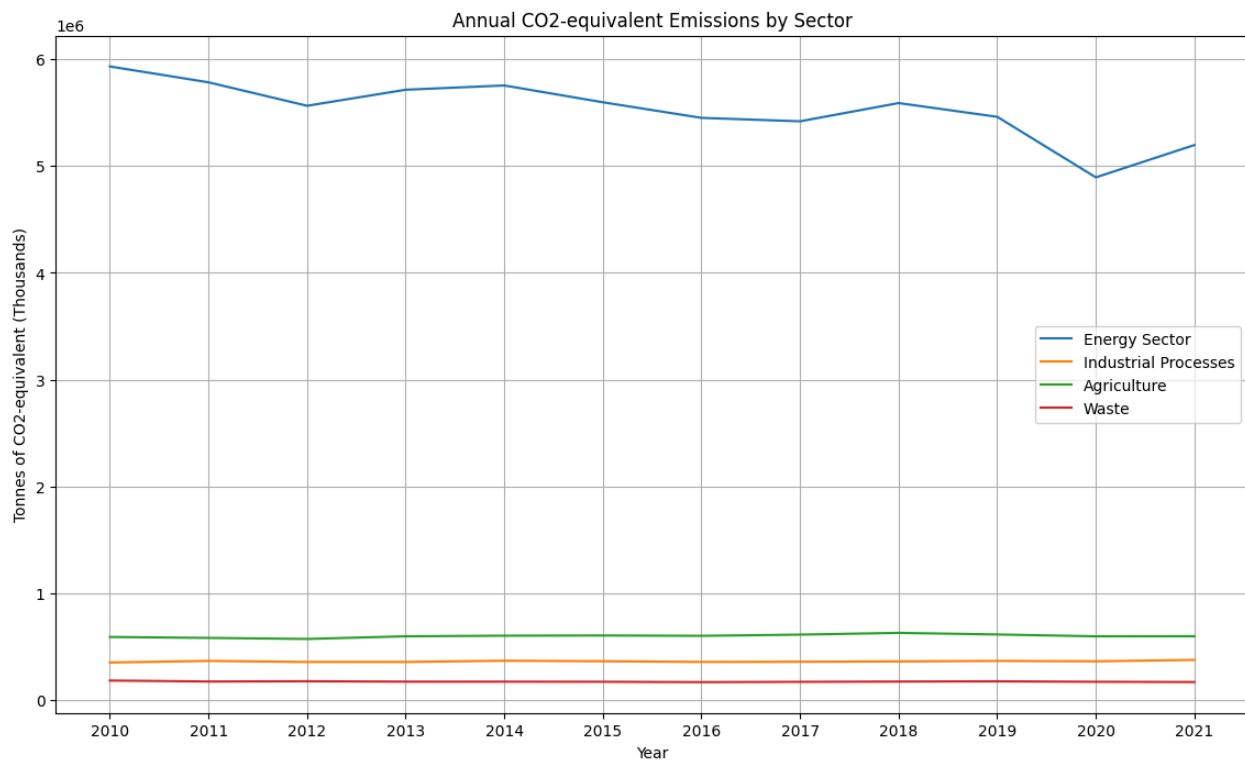
The graph shows the distribution of greenhouse gas emissions by sector, indicating that the Energy Sector is responsible for the vast majority, 82.9%. This was obtained by taking the mean of all the percentages through the 10 years of data I retrieved.

Agriculture contributes 9.0%, Industrial Processes 5.5%, and Waste 2.6%. This data supports the idea that the Energy Sector is a key area for intervention. It also suggests that while other sectors contribute, focusing on energy could yield the most significant impact on overall emissions reduction as it is a large percentage of greenhouse gas emissions. It is important to note that while not included here, the energy sector includes energy, manufacturing, and construction industries, transportation, fuel emissions, residential, and other source of energy.

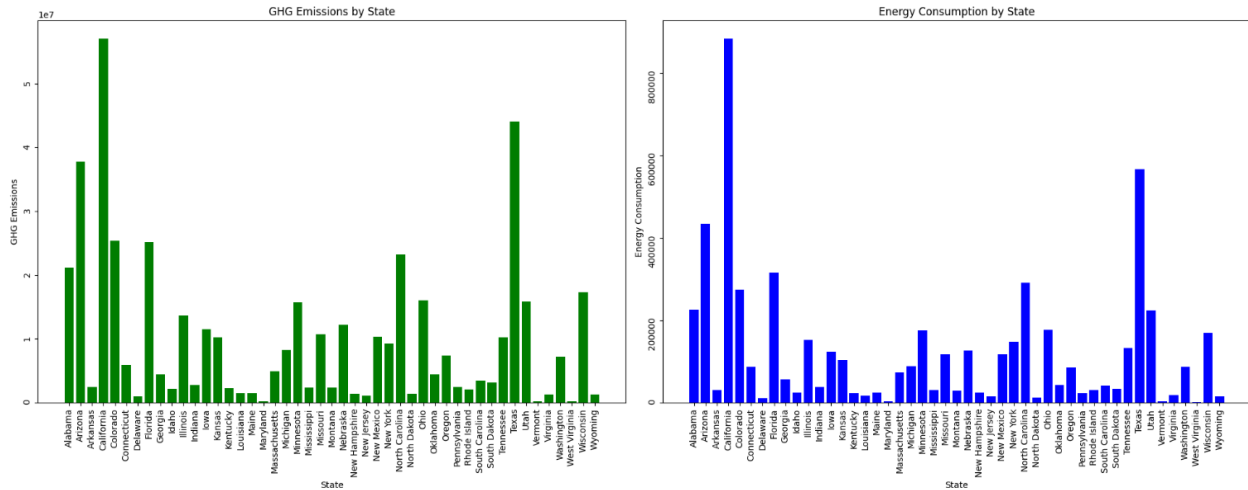


The time series graph, which uses the same data frame from the pie chart, shows the progressive growth of carbon dioxide emissions by the same sectors as measured in tonnes. The graph suggests that while emissions from agriculture, industrial processes,

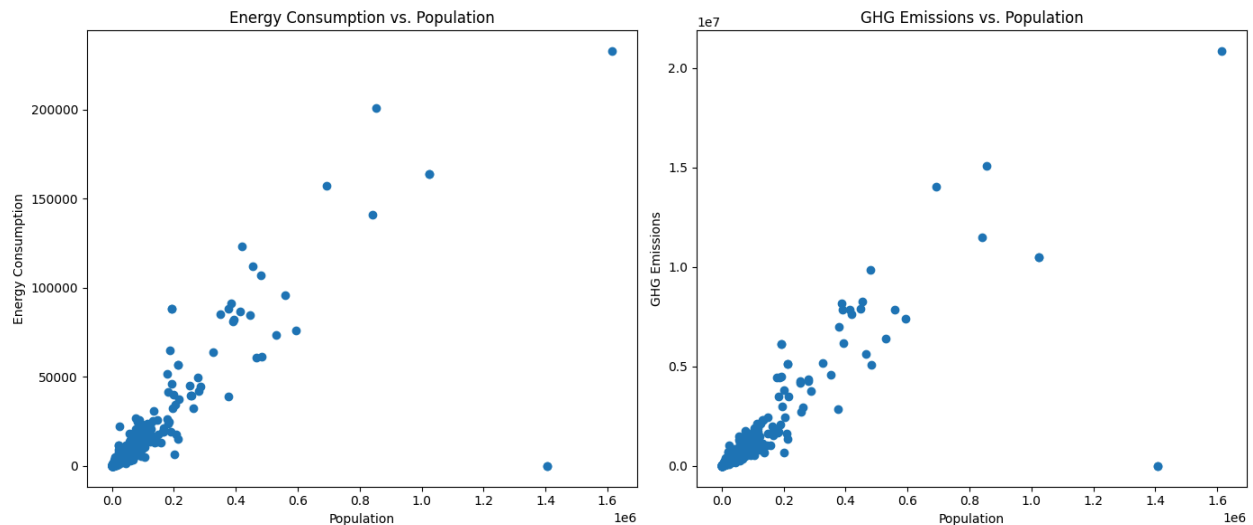
and waste have remained relatively constant, emissions from the energy sector, though initially high, show fluctuations with going down since 2010-2020 but starting to increase again in 2020-2021. This could be representative of some policies or awareness towards the impact of energy use that would lead it to go down during those times.



The plot displays two bar graphs side by side, one representing Greenhouse Gas (GHG) Emissions and the other showing Energy Consumption by state. The GHG emissions graph (green) and the Energy Consumption graph (blue) illustrate differences among states. Generally, states with higher energy consumption also show higher GHG emissions, suggesting a correlation between energy use and environmental impact. However, the direct relationship can vary by state due to factors like energy sources and efficiency measures.



The two plots illustrate the relationship between population and both energy consumption and GHG emissions. Both show a trend where, generally, larger populations are associated with higher energy usage and greater emissions. This supports the hypothesis that more populous areas contribute more to environmental degradation. Analysis of these plots can provide insights into which regions may require more stringent energy and emissions policies.



The combined analysis of the four plots provides a comprehensive view of the environmental impact across the United States. The AQI map shows that while many

areas have good air quality, certain regions, likely urban or industrial, exhibit higher pollution levels. The pie chart demonstrates the predominant contribution of the energy sector to greenhouse gas emissions, suggesting a significant area for policy focus. The time series underlines fluctuations in the energy sector's emissions, hinting at the impact of regulatory or technological changes. Meanwhile, the scatter plots and bar graphs correlate higher population and energy consumption with increased GHG emissions, highlighting the need for targeted environmental policies in more populous and energy-intensive states. This analysis underscores the intertwined nature of population, energy usage, and environmental impact, guiding where interventions might be most effective.

Conclusion

This report culminates a comprehensive examination of the interconnections between population density, energy consumption, and greenhouse gas emissions within the United States. From data acquisition through meticulous processing and insightful visual analysis, the findings corroborate the initial hypothesis, underscoring the significant environmental impact of densely populated areas due to heightened energy demands. Through various methodologies, including API data retrieval, reverse geocoding and data scraping, we've navigated through challenges to illuminate the critical role of the energy sector and urban density in contributing to emissions. The visualizations provided offer a clear depiction of how these factors interplay across different states and cities, reinforcing the urgency for targeted sustainable interventions. This analysis not only validates the proposed hypothesis but also sets a foundation for policy reform, emphasizing the necessity of a collaborative approach towards sustainable energy

utilization and emissions reduction to ensure a healthier environment for future generations.